

Chemistry V2 — ExamSIDE Solution & Transfer Book

CHEMISTRY V2 | 1

Try every question on your own first. If you get stuck there are three nudges: what to notice, what to use, and how to start. The answer check and the full worked answer are on the next page, so you can check yourself without reading the whole solution.

Question 1

Where this question comes from: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Main idea being tested: Ionic charge notation

Also uses: Formula anatomy

Where you learned this: Separate ionic charge from subscripts; Read a chemical formula before calculating

Try it first

Attempt the original question independently before opening any hint or solution.

In the ion SO_4^{2-} , what does the superscript $2-$ represent?

- A. The number of oxygen atoms
- B. The overall ionic charge
- C. A coefficient for the species
- D. The number of sulfur atoms

Your working space

- target notation = _____
- subscript(s) = _____
- superscript charge = _____
- meaning = _____

First nudge — what to notice

Notice where the numeral is written: upper right or lower right?

Second nudge — what to use

A superscript at the upper right gives the overall ionic charge; a subscript counts atoms within the formula.

Third nudge — how to start

Circle the superscript $2-$ and label it 'species charge' before evaluating the options.

The path through this question

- Read the given representation exactly as written or shown.
- Identify the chemical entities that must be tracked.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Classify only after the decisive evidence is established.
- Check that the final conclusion follows from the evidence and rule used.

Question 1 of 5. Try it yourself before you open any of the nudges.

Anatomy of the formula

SUBSCRIPT 4

atoms inside one species



IONIC CHARGE $2-$

net charge on the whole ion

Notation position and its local chemical meaning.

Question 1 — Full worked solution

Answer check — cover this until you finish

Correct option: B — The overall ionic charge

Full worked solution

- Read the ion exactly as written and separate the subscript from the superscript.
- Identify 4 as the oxygen-count subscript and 2^- as the overall ionic charge.
- Compare that meaning with every option before selecting the answer.

Check your answer

- Check that the same chemical species is being tracked.
- Check that the overall charge is preserved or interpreted correctly.

Final answer: B

The overall ionic charge

No additional condition or exception is recorded for this source item.

Question 2

Where this question comes from: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Main idea being tested: Particle-to-symbol translation

Also uses: none

Where you learned this: Translate between particles and symbols; Read a chemical formula before calculating

Try it first

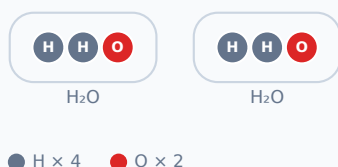
Attempt the original question independently before opening any hint or solution.

A particle diagram shows two separate H_2O particles. Which symbolic representation matches the diagram?

- A. H_2O
- B. $2\text{H}_2\text{O}$
- C. H_4O_2
- D. $2\text{H}_2 + \text{O}_2$

The picture that comes with this question

School-level particle-count model — composition and count only.



Your working space

- particle identity = _____
- number of separate particles = _____
- symbol for one particle = _____
- combined symbolic form = _____

First nudge — what to notice

Count the separate H_2O particles in the diagram before choosing a symbolic form.

Second nudge — what to use

A leading coefficient counts whole particles or formula units; changing a subscript changes the composition of one particle.

Third nudge — how to start

Write H_2O once for each drawn particle, then combine identical particles with one leading coefficient.

The path through this question

- Identify whether the information is macroscopic, particulate or symbolic.
- Identify the chemical entities that must be tracked.
- Translate one representation into another without changing identity or count.
- State what the representation means chemically.
- Check that the final conclusion follows from the evidence and rule used.

Question 2 of 5. Try it yourself before you open any of the nudges.

Particle view of the same substance

School-level particle-count model — composition and count only.



● H × 4 ● O × 2

2 particle(s) shown

Question 2 — Full worked solution

Answer check — cover this until you finish

Correct option: B — $2\text{H}_2\text{O}$

Full worked solution

- Identify each drawn particle as H_2O without changing its internal composition.
- Count two separate H_2O particles.
- Represent the count with a leading coefficient, giving $2\text{H}_2\text{O}$ rather than changing subscripts.

Check your answer

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: B

$2\text{H}_2\text{O}$

No additional condition or exception is recorded for this source item.

Question 3

Where this question comes from: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Main idea being tested: Atom conservation

Also uses: none

Where you learned this: Check atom conservation explicitly; Read a chemical formula before calculating

Try it first

Attempt the original question independently before opening any hint or solution.

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, which statement correctly verifies atom conservation?

- A. There are 2 H and 1 O atoms on each side.
- B. There are 4 H and 2 O atoms on each side.
- C. Only hydrogen is conserved.
- D. The coefficients do not affect atom counts.

Your working space

- left-side atom counts = _____
- right-side atom counts = _____
- first mismatch, if any = _____
- conservation conclusion = _____

First nudge — what to notice

Count H and O atoms separately on both sides of the equation.

Second nudge — what to use

A coefficient multiplies every atom in the formula that follows it.

Third nudge — how to start

Make a two-row ledger for H and O and calculate the left-side and right-side counts before reading the options.

The path through this question

- Read coefficients, subscripts, charges and sides in their correct positions.
- Identify the chemical entities that must be tracked.
- Apply the required conservation check.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Question 3 of 5. Try it yourself before you open any of the nudges.

Atom ledger before and after



Element	Before	After	Matched?
H	4 	4 	✓ same
O	2 	2 	✓ same

Fill or compare the required ledger and locate the first mismatch.

Question 3 — Full worked solution

Answer check — cover this until you finish

Correct option: B — There are 4 H and 2 O atoms on each side.

Full worked solution

- Apply each coefficient to the complete formula that follows it.
- Count 4 H and 2 O atoms on the reactant side.
- Count 4 H and 2 O atoms on the product side and compare the ledgers.

Check your answer

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: B

There are 4 H and 2 O atoms on each side.

No additional condition or exception is recorded for this source item.

Question 4

Where this question comes from: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 2

Main idea being tested: Reaction conditions

Also uses: none

Where you learned this: Keep reaction conditions attached

Try it first

Attempt the original question independently before opening any hint or solution.

The source represents $A \rightarrow B$ with heat written over the reaction arrow. Which restatement preserves the source condition?

- A. A forms B with the recorded heat condition retained.
- B. A forms B and the heat condition may be omitted.
- C. Heat is a product of the reaction.
- D. Heat is a coefficient multiplying A.

Condition given in the question: heat

Your working space

- reaction = _____
- recorded condition = _____
- where the condition belongs = _____
- condition-preserving statement = _____

First nudge — what to notice

Inspect the annotation attached to the reaction arrow; it is part of the source meaning.

Second nudge — what to use

A condition written over or with a reaction arrow is neither a reactant nor a product and must stay attached to the reaction description.

Third nudge — how to start

Copy the reaction meaning while explicitly retaining the word 'heat' as the recorded condition.

The path through this question

- Read the given representation exactly as written or shown.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Check condition or exception.
- State what the representation means chemically.
- Check that every stated condition or exception is still attached.

Question 4 of 5. Try it yourself before you open any of the nudges.

Condition gate before the rule applies



The recorded condition stays attached to the arrow.



YES

Keep it attached and apply the version it requires.

NO

Apply the default rule with nothing added.

Question 4 — Full worked solution

Answer check — cover this until you finish

Correct option: A — A forms B with the recorded heat condition retained.

Full worked solution

- Read the reaction and the arrow annotation as one source representation.
- Identify heat as a recorded reaction condition, not as a reactant, product or coefficient.
- Choose the statement that retains the heat condition with the reaction.

Check your answer

- Check that every stated condition or exception is still attached.

Final answer: A

A forms B with the recorded heat condition retained.

Preserve and verify the recorded condition: heat.

Question 5

Where this question comes from: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 2

Main idea being tested: Reacting-species roles

Also uses: none

Where you learned this: Attach a role only after tracking the species; Track the same reacting species

Try it first

Attempt the original question independently before opening any hint or solution.

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, the half-change $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is shown. Which reactant is the reducing agent?

- A. Zn
- B. Cu^{2+}
- C. Zn^{2+}
- D. Cu

Your working space

- species A: _____ $\rightarrow$ _____
- species B: _____ $\rightarrow$ _____
- electron-losing species = _____
- electron-gaining species = _____
- reducing agent = _____
- oxidising agent = _____

First nudge — what to notice

Use the shown half-change and look at which side contains electrons.

Second nudge — what to use

A reactant that loses electrons is oxidised and acts as the reducing agent because it supplies electrons to the other species.

Third nudge — how to start

From $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$, mark Zn as the electron-losing reactant before assigning the agent name.

The path through this question

- Identify the chemical entities that must be tracked.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Track the same species or state from before to after.
- Classify only after the decisive evidence is established.
- State what the representation means chemically.
- Check that the final conclusion follows from the evidence and rule used.

Question 5 of 5. Try it yourself before you open any of the nudges.

Sort the species before naming any role



Reactant species	Zn Cu²⁺
Species that changes	fill only from the evidence above
Spectator (if present)	fill only from the evidence above
Product species	Zn²⁺ Cu

Attach a role only after identifying the changing reactant species.

Question 5 — Full worked solution

Answer check — cover this until you finish

Correct option: A — Zn

Full worked solution

- Track the reactant species through the shown half-change.
- Zn appears on the reactant side and produces electrons, so Zn loses electrons and is oxidised.
- The electron-losing reactant is the reducing agent; attach the role to Zn only after that change is established.

Check your answer

- Check that the same chemical species is being tracked.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: A

Zn

No additional condition or exception is recorded for this source item.